AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 13 · STUDENT LESSON

Train a Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 11: selecting a threshold is different from applying it.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 19

Your mission

> Start with the target: Connect training-only scaling and neighbour prediction in one pipeline. Try the recall before opening notes.

Your learning target

I can connect **training**-only scaling and neighbour **prediction** in one **pipeline**.

I can **reproduce predictions** from recorded data settings and fitted values.

I can audit a **pipeline** for re**fitting** feature-order errors and evaluation leakage.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Training fits model **parameter**s from labeled examples; **prediction** applies the frozen fitted model to new feature rows.



Your target

LEARNING TARGET **Training**: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

Builder name and experiment date:

Readiness for the connected explanation

Which values does min-max scaling learn, and which values should be reused for a validation query?

Readiness for Searching candidate rules

How does a **training**-majority baseline choose its **prediction**, and where is its validation accuracy measured?

My readiness explanation

This week's end goal

By the end, I can explain train a classifier using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Train a Classifier case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use train a classifier to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A **pipeline** connects steps so a new query receives the same preparation as the **training** examples. This week combines the scale from Week 11 with a one-neighbour classifier.

We will fit once, keep the **fitted state**, and call one **prediction** function for every raw query.

Use the same fictional trip data. Each feature pair is ordered (distance in cm, brightness in sensor units). **Training** IDs are A,B,C.

Validation IDs V1,V2,V3 belong to separate trips; related trips were grouped before splitting. Use min-max scaling, sum of absolute scaled differences, k=1 and alphabetical ID order for exact distance ties.

Training ID	Feature pair	Label
A	(10,200)	clear
B	(20,100)	blocked
C	(20,200)	blocked

The **input contract** requires two finite numeric features in the stated order and units, with no missing values. **Training** also needs unique IDs, known labels, at least one row and nonzero ranges.

These assumptions must be checked before this short classroom program is used; its explicit guard only checks zero ranges.

Apply the connected idea: Searching candidate rules

Last week the baseline ignored scene features. This week a small **fitting** algorithm will choose a rule that uses one feature.

We supply the rule form: **predict** blocked when a distance reading is at or below a cutoff, otherwise **predict** clear. The algorithm will try three cutoffs and keep the one making the fewest **training** mistakes.

The teacher-authored fictional rows contain a sensor distance in centimetres and a separate reviewed lane label, blocked or clear. The reading is available before **prediction**; the label comes from the scene review.

A single distance need not describe every obstruction. These numbers are for learning, not real robot control. Each row comes from a different trip, and **training** and validation trips are separate.

Training row	Distance in cm	Reference label
1	2	blocked
2	4	blocked
3	6	clear
4	8	clear

Before scoring, declare cutoffs 3, 5 and 7 cm in ascending order, **predict** blocked at equality, and keep the smaller cutoff if **training error**s tie. This search tests only these three rules with this direction.

It does not find the best possible model or prove that distance causes the label.

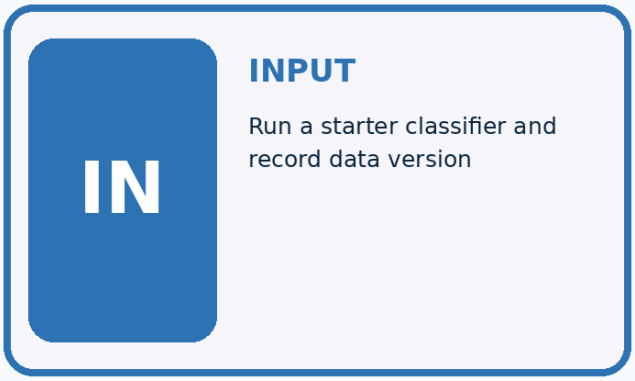
See how it works

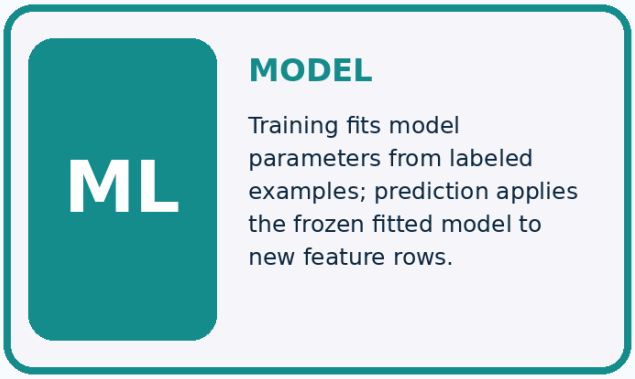
> Follow the mechanism for Train a Classifier. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Run a starter classifier and record data version, split, settings, and **prediction**s.

MODEL IDEA **Training** fits model **parameter**s from labeled examples; **prediction** applies the frozen fitted model to new feature rows.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Load data version v1 and a saved train/test split. Fit pre**process**ing on **training** rows only, then fit the classifier.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

```
def scale(x, low, span):
    return tuple((x[j] - low[j]) / span[j] for j in range(2))

def fit(rows):
    xs = [x for item_id, x, label in rows]
    low = tuple(min(x[j] for x in xs) for j in range(2))
    span = tuple(max(x[j] for x in xs) - low[j]
                 for j in range(2))
```

```
    if 0 in span:
        raise ValueError("Review a constant training feature")
    stored = tuple((i, scale(x, low, span), y) for i, x, y in rows)
    return low, span, stored

def predict(x, model):
    low, span, stored = model
    q = scale(x, low, span)
    ranked = sorted((sum(abs(a - b) for a, b in zip(q, v)), i, y)
                    for i, v, y in stored)
    return ranked[0][1], ranked[0][2]

train = [("A", (10, 200), "clear"),
         ("B", (20, 100), "blocked"),
         ("C", (20, 200), "blocked")]
model = fit(train)
queries = [(11, 120), (19, 190), (12, 80)]
for x in queries:
    print(predict(x, model))
```

fit returns minima (10,100), ranges (10,100) and scaled **training** examples. **predict** transforms a raw query with that state, ranks distances, then returns the selected ID and label. zip pairs matching features; sum adds their absolute differences. sorted uses distance first and unique ID second.

Run the whole script in Python 3. It prints ('A', 'clear'), ('C', 'blocked'), ('B', 'blocked') on separate lines. No validation labels enter fit or **predict**.

This is a small **pipeline** written with functions, not the scikit-learn **Pipeline** class.

Apply the connected idea: Searching candidate rules

```
train = [(2, "blocked"), (4, "blocked"),
         (6, "clear"), (8, "clear")]
best_errors = len(train) + 1
best_cutoff = None
for cutoff in [3, 5, 7]:
    errors = 0
    for distance, label in train:
        guess = "clear"
        if distance <= cutoff:
            guess = "blocked"
        if guess != label:
            errors = errors + 1
    print(cutoff, errors)
```

```
    if errors < best_errors:
        best_errors = errors
        best_cutoff = cutoff
print("chosen", best_cutoff)
```

The **output** lines are 3 1, 5 0, 7 1, then chosen 5. Cutoff 3 misses the blocked row at 4 cm. Cutoff 5 matches all four **training** labels. Cutoff 7 wrongly marks the 6 cm clear row as blocked.

Fitting stores cutoff 5; **prediction** then applies that stored rule to a new distance.

Separate search from later scoring

STAGE	EVIDENCE	DECISION
Try cutoffs 3 5 7	Training labels	Count 1 0 1 errors
Store cutoff 5	Smallest training error	Freeze this rule
Compare with baseline	Same validation rows	Report both scores

The supplied search chooses a cutoff using training errors. A perfect training fit does not establish performance on new examples.

best_errors starts above any possible mistake count. The strict < keeps the first tied candidate; ascending order makes that the smaller cutoff.

The code assumes nonempty **training** data, numeric distances and the two allowed labels. Validate those assumptions before using another dataset.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 19

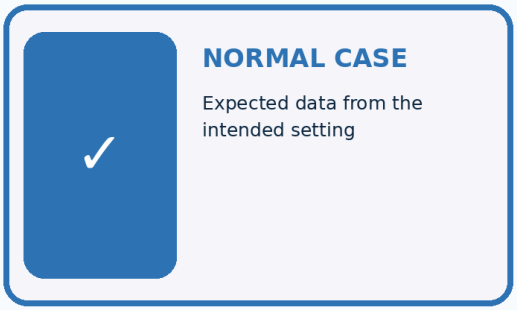
Compare the cases

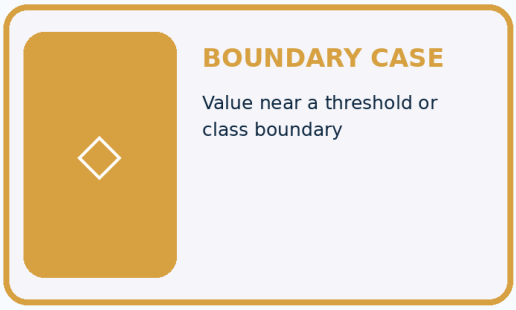
> Compare the supplied cases for Train a Classifier. Identify the detail that changes the conclusion.

EVALUATION DESIGN

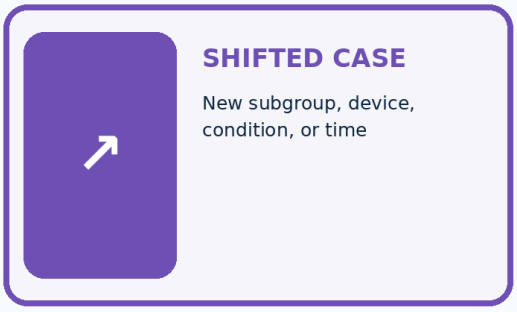
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Transform **training** but leave query numerically at 5.

Predicted result

Prediction: Distances become 15 and 55, incorrectly selecting healthy.

Why the result follows

Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

STUDENT LESSON · 5 OF 19

Words that unlock the idea

> Use the vocabulary of Train a Classifier to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

FIT

learn parameters from training data

2

PARAMETER

learned numeric model value

3

PREDICT

apply fitted model to new features

4

PIPELINE

ordered preprocessing and model steps

Term	Precise meaning	My ML example
fit	learn parameters from training data	
parameter	learned numeric model value	
predict	apply fitted model to new features	
pipeline	ordered preprocessing and model steps	

MISCONCEPTION TO REPAIR The model keeps learning whenever it sees test examples during evaluation.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Pipeline	An ordered sequence of data transformations and prediction steps
Fitted state	The learned values and stored examples needed to make predictions
Reproduce	Obtain the same result using the same recorded inputs and procedure
Input contract	The documented feature names order units and accepted values

Terms for Searching candidate rules

Term	Meaning
Candidate rule	One possible prediction rule in a declared search
Threshold	A numerical boundary used to choose between outcomes
Training error	A mismatch between a prediction and its reference label in training data

Term	Meaning
Fitting	Using training examples to choose a model value or rule

STUDENT LESSON · 6 OF 19

Follow a worked example

> Follow the worked trace for Train a Classifier, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 13 • WORKED EVIDENCE

1. Load data version v1 and a saved train/test split.
2. Fit preprocessing on training rows only, then fit the classifier.
3. Freeze the pipeline and predict twenty test rows once.
4. Save settings, predictions, and metrics so the experiment can be reproduced.

Step 1

STEP 1 Load data version v1 and a saved train/test split.

Step 2

STEP 2 Fit pre**process**ing on **training** rows only, then fit the classifier.

Step 3

STEP 3 Freeze the **pipeline** and **predict** twenty test rows once.

Step 4

STEP 4 Save settings, **prediction**s, and metrics so the experiment can be **reproduce**d.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Two stages with one fitted state

STAGE	INPUT AND ACTION	RESULT
Fit once	Training rows only	Scale plus stored rows
Predict each query	Raw pair plus state	Neighbour ID and label
Evaluate afterward	Prediction plus reference	Recorded match or error

Reference labels are used after prediction to score results. They do not enter the fitted scale or stored neighbour pool.

Worked explanation · Train a Classifier

For raw query (11,120), scale gives (0.1,0.2). Distances to A,B,C are 0.9,1.1,1.7, so the function returns A and clear. It does not recalculate minima or add the query to stored rows.

The returned ID is useful for checking the reasoning, but it is not **evidence** that the label is true.

Trace (11,120) through both functions. What does fit store, what does predict reuse, and at which point may a validation reference label be compared?

Store the **fitted state** together with the **input contract** and method settings. A model without its scale or feature order is incomplete.

Swapping the raw pair to (120,11) produces B and blocked here; the function cannot detect that the caller swapped the meanings.

Apply the connected idea: Searching candidate rules

Now reveal four validation rows in order: (3, blocked), (5, clear), (7, clear), (9, blocked). With the stored cutoff 5, **prediction**s are blocked, blocked, clear, clear.

Rows 1 and 3 match, so validation accuracy is 2/4 or 50 percent despite zero **training** mistakes.

Worked explanation · Train a Classifier

The **training** labels tie at two blocked and two clear. Carry forward Week 7's baseline tie rule, choose clear. That baseline matches validation rows 2 and 3, also 2/4 or 50 percent.

The fitted rule has not improved total accuracy on this set, and the two methods make different mistakes.

For the validation distance exactly 5 cm, what does the fitted rule predict and why? Which rows are wrong for the rule and the baseline? State their difference in percentage points.

Worked explanation · Train a Classifier

Keep the rule and baseline fixed for this comparison. Do not secretly choose a different cutoff after seeing validation labels and call that a **training**-only choice.

Validation can guide later development, but record those choices and reserve an untouched final test. No final test is supplied here.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Train a Classifier

Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

The facts we will use

Supplied facts

Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance. Query is 5 cm.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/13

Train a Classifier

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fit transform

The declared conversion produces training lengths 20 and 60 mm.

2. Transform query

5 cm becomes 50 mm using the same conversion.

3. Predict neighbour

Distances 30 and 10 mm select the spotted training record.

CHANGE ONE THING

Transform training but leave query numerically at 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit transform	The declared conversion produces training lengths 20 and 60 mm.
2. Transform query	5 cm becomes 50 mm using the same conversion.
3. Predict neighbour	Distances 30 and 10 mm select the spotted training record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Distances 30 and 10 mm select the spotted **training** record. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records

and appropriate group splits.

Keep the stated limit

Read the labelled picture: Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Transform **training** but leave query numerically at 5. **Predict** the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 19

Find and repair a mistake

> Find the unsupported step in the Train a Classifier claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The model keeps learning whenever it sees test examples during evaluation. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE The model keeps learning whenever it sees test examples during evaluation.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

Check that fit receives **training** rows only, **predict** receives raw features in the correct order, and the **fitted state** stays unchanged during **prediction**.

Do not scale a query twice, fit again on validation, or add labelled validation rows to the neighbour pool. A **pipeline** still leaks if the wrong rows are passed to its fit step.

The tuple state groups the learned numbers and stored examples. Keep the associated data and code versions with it, and compare stored values and **output**s when repeating a run.

Do not accept a matching score alone: two different **prediction** lists can have the same score.

A learner gets the same 2/3 score twice after fitting on training plus validation rows. They claim the pipeline must be valid because the result repeats. Explain why the claim fails and describe the repair.

Apply the connected idea: Searching candidate rules

Check the claim against the facts

Humans chose the feature, labels, candidate cutoffs, direction and scoring method. The program used **training** examples to select the cutoff.

That is a small example of **fitting** a model; writing a loop alone does not make a method learn. Choosing the cutoff in advance without consulting **training** examples would instead be a fixed rule.

Check the claim against the facts

Record the source and split, rule form, candidate list, all **training error** counts, tie policy, selected cutoff and both validation **prediction** lists. Do not discard a difficult row to improve a score.

Next week a decision tree will extend the idea of feature-based decisions, while the need for honest evaluation stays the same.

A learner reports that cutoff 5 is perfect because training error is zero. Rewrite the claim using the training result, validation comparison and the limits of this small search.

STUDENT LESSON · 8 OF 19

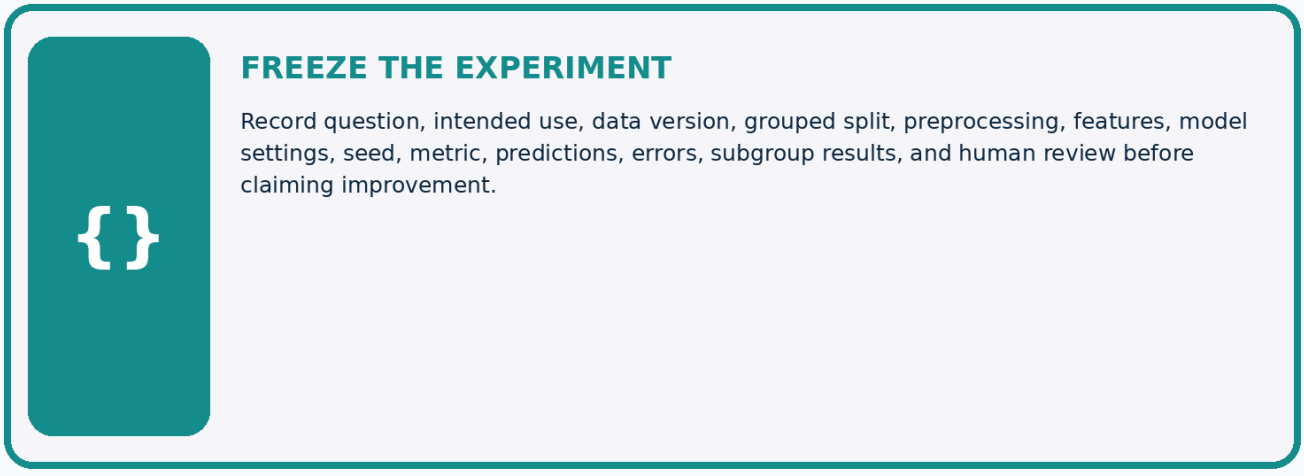
Practise together

Practise Train a Classifier with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be **reproduce**d from the recorded data, code, settings, and evaluation rule.



Experiment record

> ### Question to explain
>
> QUESTION + intended use + success **threshold**

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to **reproduce** today's result:

A closer explanation

Record dataset and code versions, **training** IDs A B C, validation IDs V1 V2 V3, feature order and units, fitted minima and ranges, k=1, distance and tie policy, zero-range policy and the actual Python version.

Save the **prediction**s in validation ID order. This script has no random step; a seed would not fix changed data or swapped features.

References for V1,V2,V3 are clear,blocked,clear. The printed **prediction**s clear,blocked,blocked match V1 and V2: 2/3. The **training**-majority blocked baseline matches only V2: 1/3.

Rerunning the same code and **input**s should **reproduce** these **output**s, including the V3 mistake. Reproducibility does not establish correctness or performance on new data.

Changing the scaling, k or feature list creates a new development run to record. Keep a separate final test for the final evaluation. Next week a confusion matrix will show which kinds of label errors the **prediction**s make.

Apply the connected idea: Searching candidate rules

A learner uses distance < cutoff instead of distance <= cutoff during **prediction**. That changes the boundary: at 5 cm the stored cutoff now **predict**s clear. Recheck code against the declared rule before interpreting the score.

A small symbol can change which examples match.

Another learner uses <= when deciding whether a candidate has fewer errors. That replaces an earlier candidate on a tie. The **prediction** boundary <= and the search comparison < have different jobs.

Use the declared tie policy; a tied **training** score gives no **evidence** that one tied rule will be better on new cases.

Practise with less help: Trace and score the saved run

Use workbook W2 for the connected example “A reproducible classifier **pipeline**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for A reproducible classifier **pipeline**. Proposed explanation: “A **pipeline** prevents every leak automatically”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Searching candidate rules

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: Zero **training error** means perfect **prediction**s

Explain the picture

Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 19

Run the investigation

> Investigate Train a Classifier. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Starter Classifier Run: trace code, predict outputs, execute the fixed pipeline, and compare recorded results with a peer.

1. Write the hypothesis and expected **evidence** before running.

2. Starter Classifier Run: trace code, **predict output**s, execute the fixed **pipeline**, and compare recorded results with a peer.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

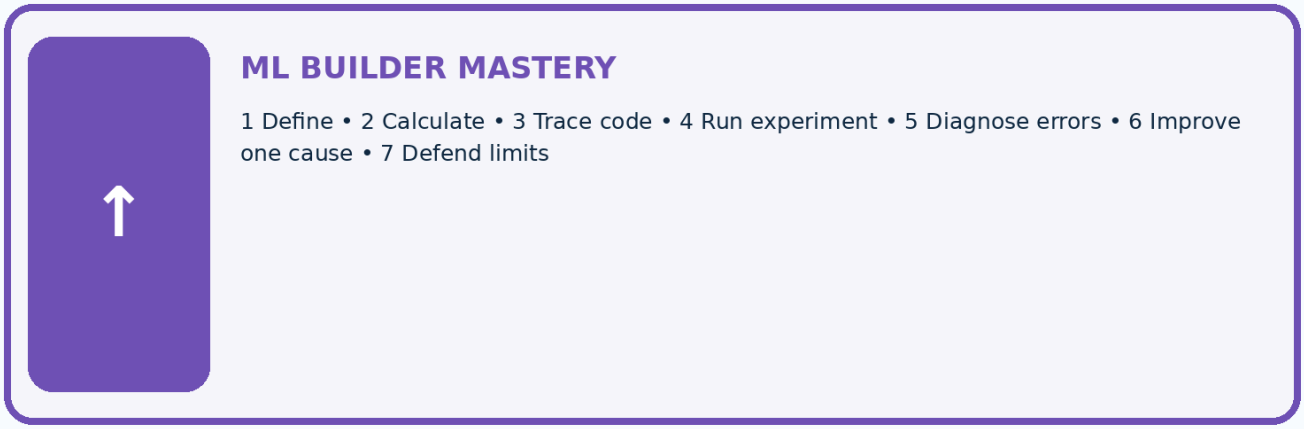
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Train a Classifier. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: fit, **parameter**, **predict**, **pipeline**.

CALCULATE **Reproduce** the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The model keeps learning whenever it sees test examples during evaluation.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 19

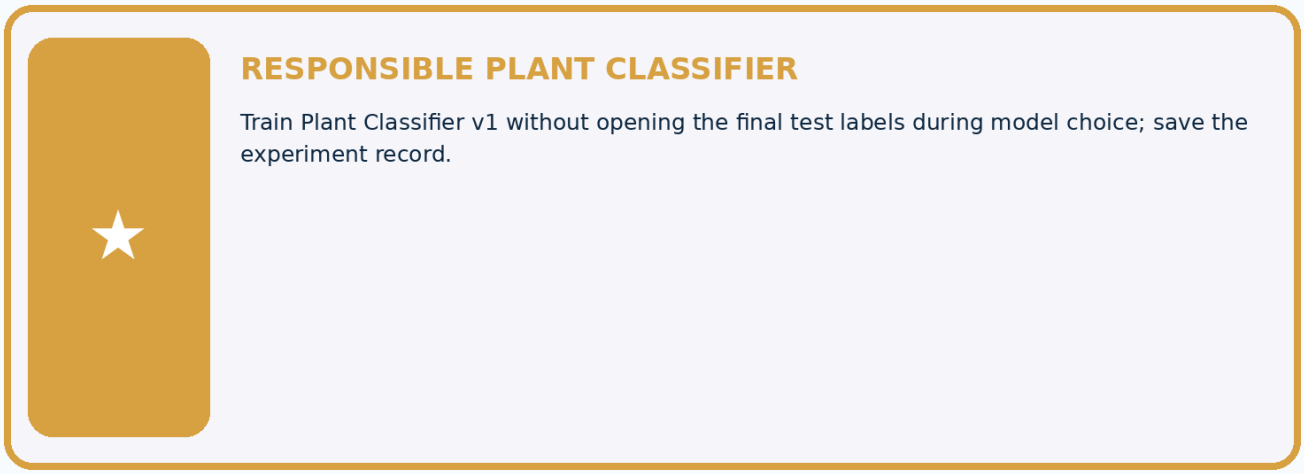
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Train Plant Classifier v1 without opening the final test labels during model choice; save the experiment record.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 19

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Run a starter classifier and record data version, split, settings, and **prediction**s.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Scores and Calibration Intuition.

STUDENT LESSON · 13 OF 19

Study a complete case

> Revisit the complete Train a Classifier case for supported practice. Repeated examples do not become fresh assessments.

Week 13 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Training is (0,H),(1,H),(4,S),(5,S). **Candidate rule predict**s S for spots>=3. Held-out cases are (2,H),(3,S).

The question

Apply the rule and distinguish **fitting** from inference.

Worked explanation

Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Apply a developed rule to new cases

Evidence or stage	Value or result
Training	0 and 1 labelled H, 4 and 5 labelled S
Supplied rule	Spots at least 3 predicts S
Test 2 spots	Predict H
Test 3 spots	Predict S

Training is (0,H),(1,H),(4,S),(5,S). **Candidate rule predict**s S for spots>=3. Held-out cases are (2,H),(3,S).

Apply the rule and distinguish **fitting** from inference.

Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

STUDENT LESSON · 14 OF 19

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Build and trace a fresh fitted state

Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new **training** rows. Fit the same **pipeline**. **Predict** validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked.

Give fitted minima and ranges, transformed queries, selected IDs, **prediction**s, accuracy and **training**-majority baseline.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 13 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classifier fits on rows 1–80 and is evaluated on those same rows. Its accuracy is 100%. What additional **evidence** is needed before claiming generalization?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. **Predict** the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Train Plant Classifier v1 without opening the final test labels during model choice; save the experiment record.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Training: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 19

Fit and test a threshold classifier

> Follow the supplied Train a Classifier lab. Predict first, keep actual results and explain the changed test.

The program chooses a **threshold** using **training** labels only. Spotted is 1 and healthy is 0. Candidate **threshold**s lie between consecutive **training** values. Test rows are used after selection.

```
"""Fit a threshold using training data, then evaluate untouched examples."""
train = [(0, 0), (1, 0), (4, 1), (5, 1)]
values = sorted({x for x, y in train})
if len(values) < 2:
    raise ValueError("Two distinct training values are required")
candidates = [(a + b) / 2 for a, b in zip(values, values[1:])]


def errors(threshold):
    return sum(int(x >= threshold) != y for x, y in train)


threshold = min(candidates, key=errors)
test = [(2, 0), (3, 1)]
predictions = [int(x >= threshold) for x, y in test]
correct = sum(pred == y for pred, (x, y) in zip(predictions, test))
print(threshold, predictions, correct, len(test))
```

Expected **output**: 2.5 [0, 1] 2 2

Candidate **threshold**s 0.5, 2.5 and 4.5 have **training error**s 1, 0 and 1. The selected **threshold** is 2.5. Both test **prediction**s are correct.

This tiny artificial example demonstrates **fitting** and evaluation; it does not establish reliability on real leaf photos.

Independent code change

Change the final test row to (3, 0), keeping **training** unchanged. **Predict** before running.

My **prediction** and **evidence**

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 16 OF 19

Your lesson route

> Use this map to navigate Train a Classifier, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

By the end, I can	How I will show it
Connect training-only scaling and neighbour prediction in one pipeline	Explain the deciding step, show a trace or test, and state one limit.
Reproduce predictions from recorded data settings and fitted values	Explain the deciding step, show a trace or test, and state one limit.
Audit a pipeline for refitting feature-order errors and evaluation leakage	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 17 OF 19

Finish and return

> Explain Train a Classifier without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Connect **training**-only scaling and neighbour **prediction** in one **pipeline**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 18 OF 19

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Distances 30 and 10 mm select the spotted **training** record.

1. **Predict** neighbour

2. Fit transform

3. Transform query

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Predict** neighbour Distances 30 and 10 mm select the spotted **training** record. Read the picture in order. Distances 30 and 10 mm select the spotted **training** record.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Transform **training** but leave query numerically at 5. Which conclusion is supported by the supplied facts?

1. Distances become 15 and 55, incorrectly selecting healthy.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Distances become 15 and 55, incorrectly selecting healthy. The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Repair a missing building block

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training** accuracy replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. **Predict** first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

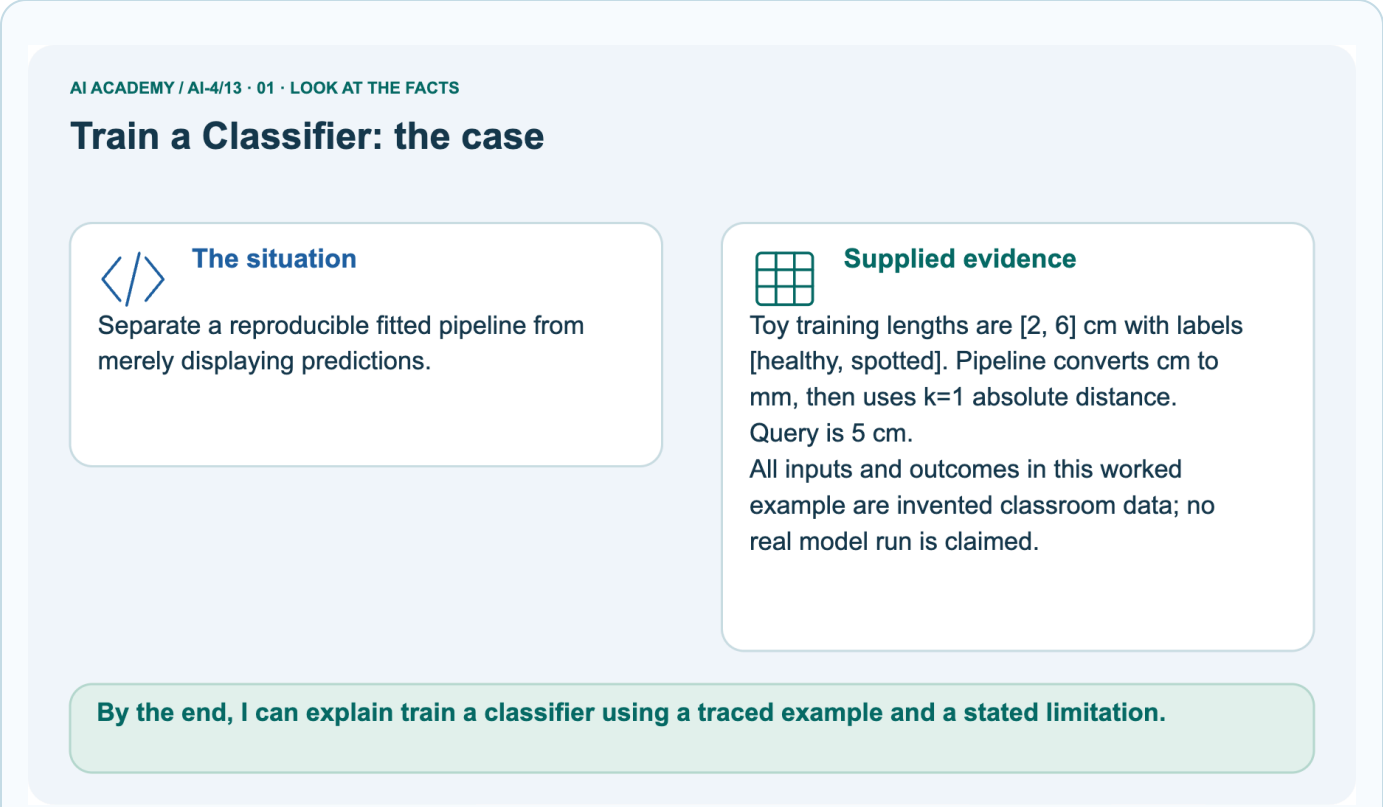
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 19 OF 19

Visual concept journey

> Explain each Train a Classifier graphic using its labels, deciding evidence and relevant limitation.

01. Train a Classifier: the case



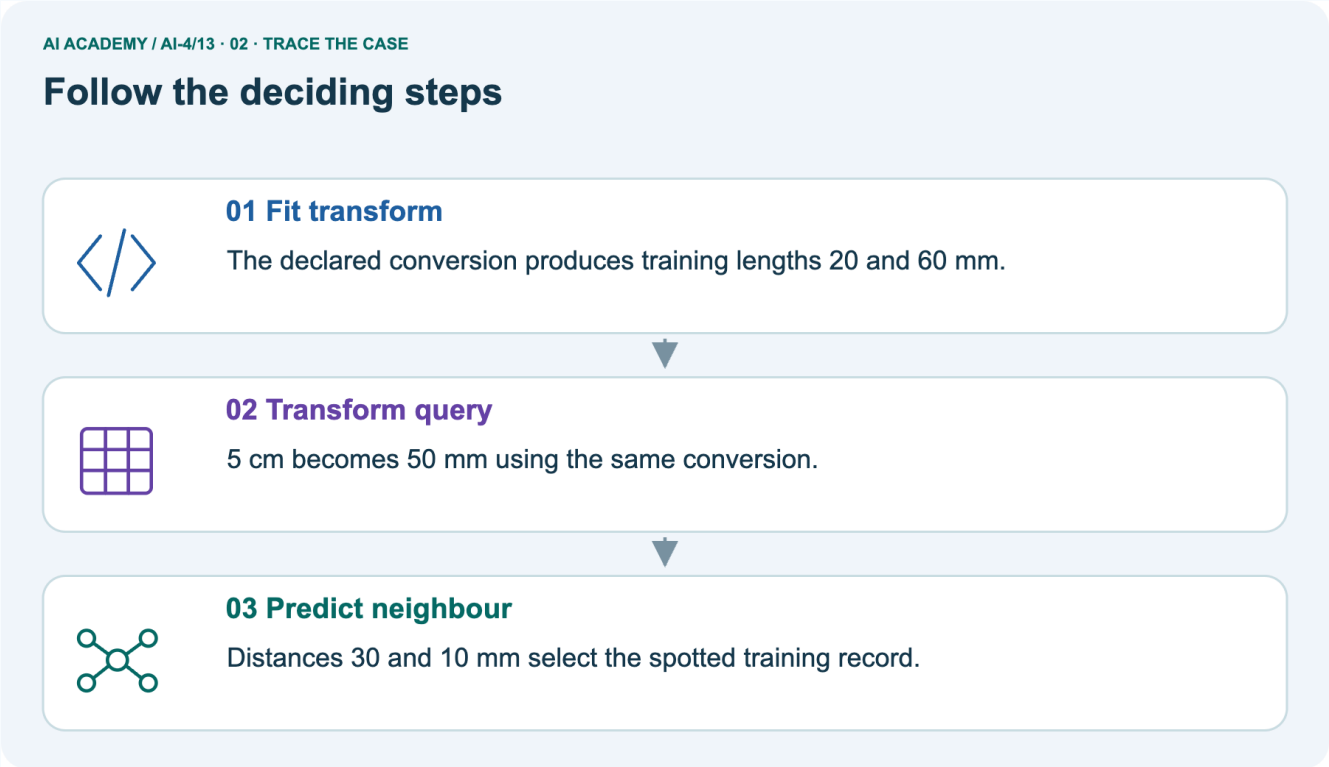
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate a reproducible fitted **pipeline** from merely displaying **prediction**s. Supplied facts: Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance.

Query is 5 cm. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

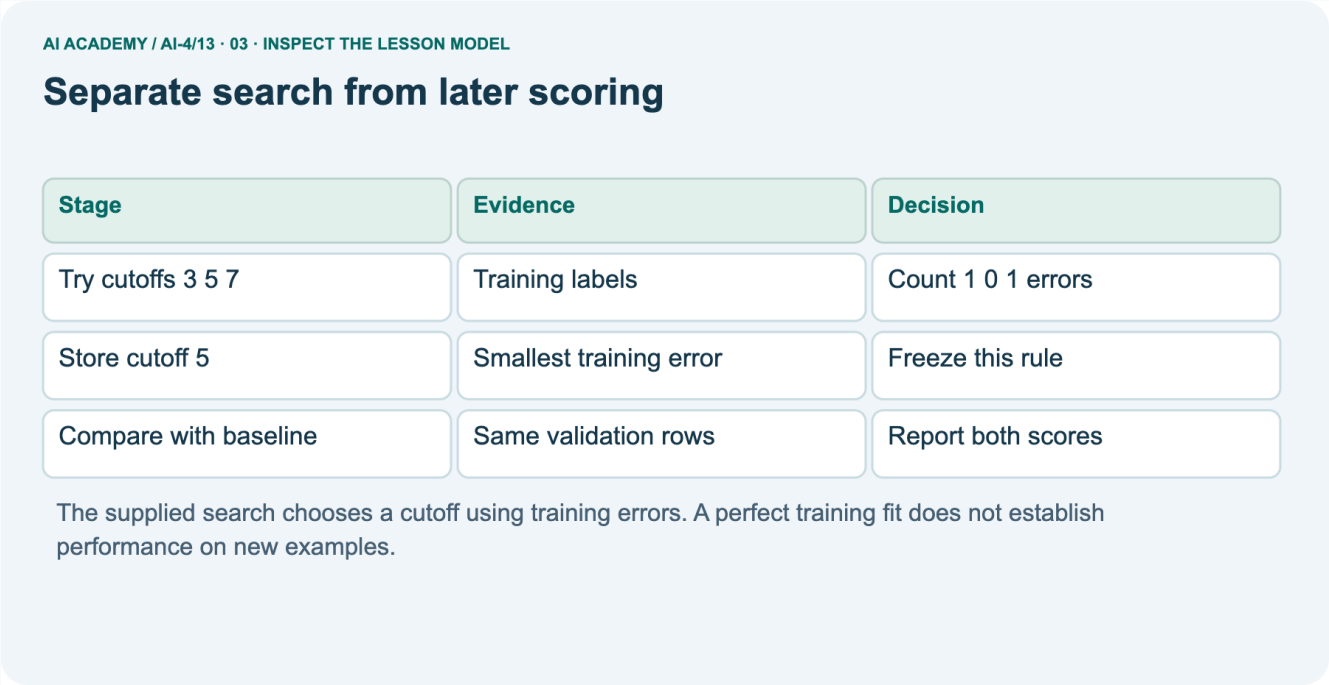


Follow the labelled picture with the matching explanation. Open the vector diagram

Fit transform: The declared conversion produces **training** lengths 20 and 60 mm. → Transform query: 5 cm becomes 50 mm using the same conversion. → **Predict** neighbour: Distances 30 and 10 mm select the spotted **training** record.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Separate search from later scoring

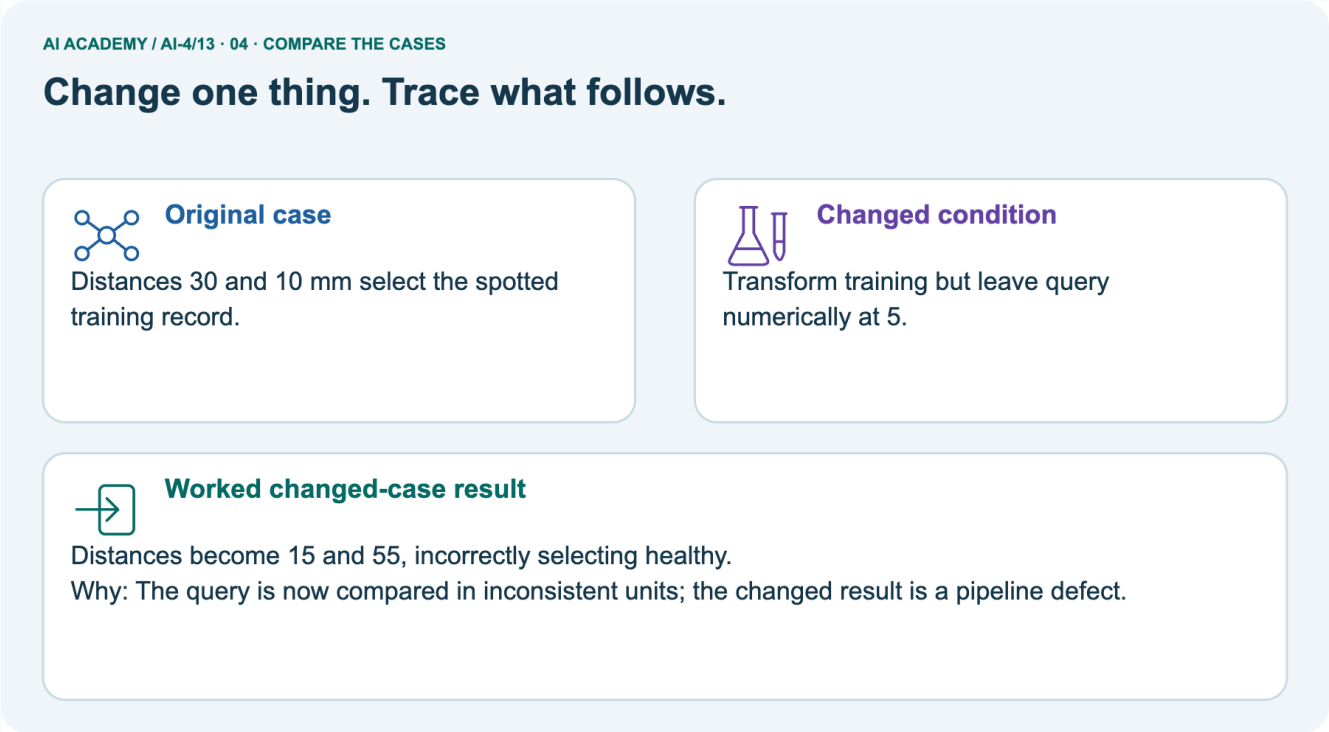


Follow the labelled picture with the matching explanation. Open the vector diagram

Separate search from later scoring. Try cutoffs 3 5 7 | **Training** labels | Count 1 0 1 errors. Store cutoff 5 | Smallest **training error** | Freeze this rule. Compare with baseline | Same validation rows | Report both scores

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



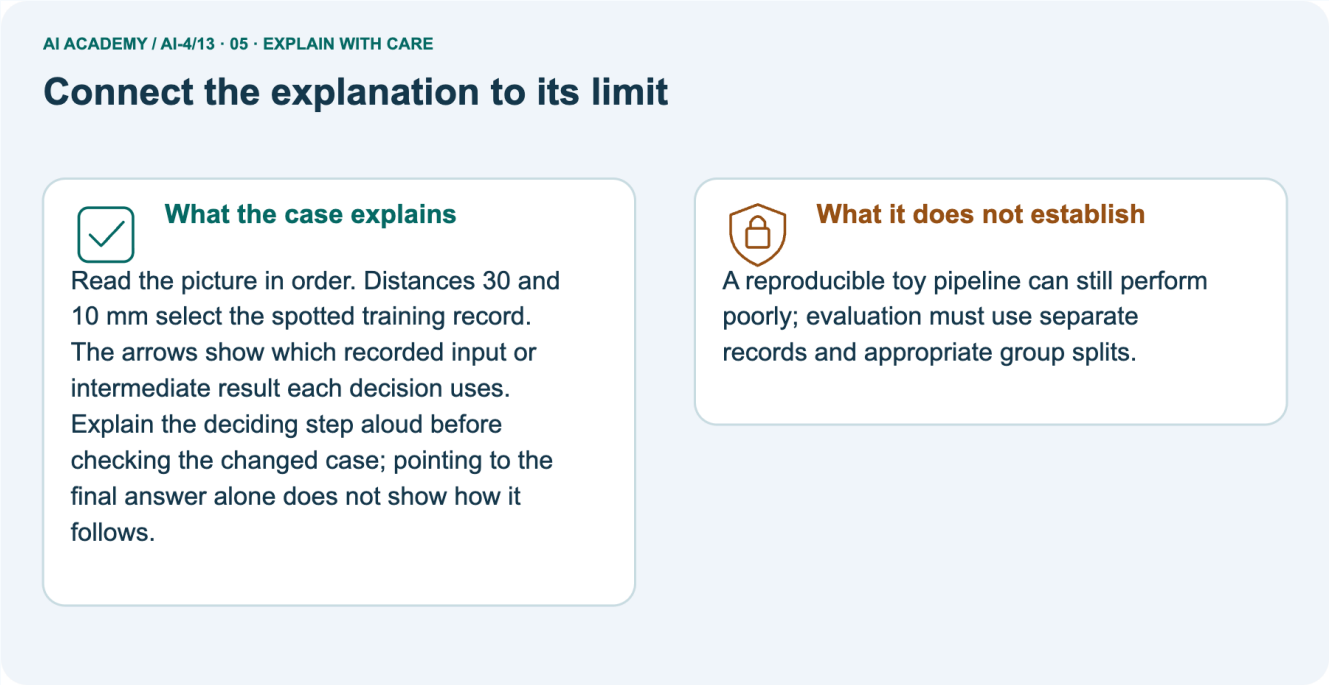
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Distances 30 and 10 mm select the spotted **training** record. Changed condition: Transform **training** but leave query numerically at 5. Expected result: Distances become 15 and 55, incorrectly selecting healthy.

Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

> **Predict** before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

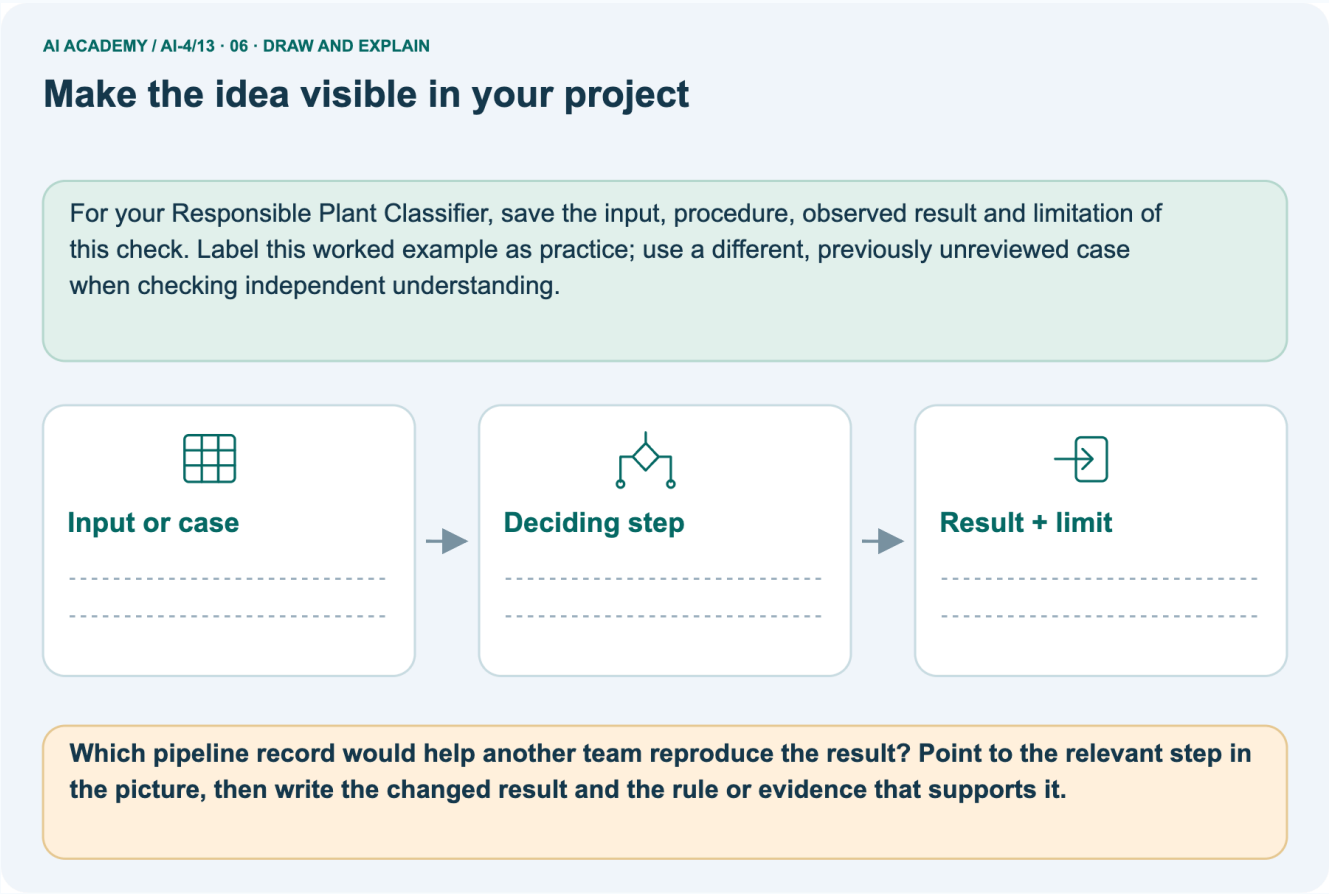
Read the picture in order. Distances 30 and 10 mm select the spotted **training** record. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records and appropriate group splits.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.